A typical termite nest is founded by a royal pair, the king and queen, who then mate with each other exclusively until one of them dies. His or her place is then taken by one of their offspring who mates incestuously with the surviving parent. If both of the original royal couple die, they are replaced by an incestuous brother–sister couple. And so on. A mature colony is likely to have lost several kings and queens, and the progeny being turned out after some years are likely to be very inbred indeed, just like laboratory rats. The average homozygosity, and the average coefficient of relatedness, within a termite nest creeps up and up as the years go by and royal reproductives are successively replaced by their offspring or their siblings. But this is only the first step in Hamilton's argument. The ingenious part comes next.

The end-product product of any social insect colony is new, winged reproductives who fly out of the parent colony, mate, and found a new colony. When these new young kings and queens mate, the chances are that these matings will *not* be incestuous. Indeed, it looks as though there are special synchronizing conventions designed to see to it that different termite nests in an area all produce winged reproductives on the same day, presumably in order to foster outbreeding. So, consider the genetic consequences of a mating between a young king from colony A and a young queen from colony B. Both are highly inbred themselves. Both are the equivalent of inbred laboratory rats. But, since they are the product of different, *independent* programs of incestuous breeding, they will be genetically different from one another. They will be like inbred white rats belonging to different laboratory strains. When they mate with each other, their offspring will be highly *heterozygous*, but *uniformly* so. Heterozygous means that at many of the genetic loci the two genes are different from each other. Uniformly heterozygous means that almost every one of the offspring will be heterozygous in exactly the same way. They will be genetically almost identical to their siblings, but at the same time they will be highly heterozygous.

Now jump forward in time. The new colony with its founding royal pair has grown. It has become peopled by a large number of identically heterozygous young termites. Think about what will happen when one or both of the founding royal pair dies. That old incest cycle will begin again, with remarkable consequences. The first incestuously produced generation will be dramatically more variable than the previous generation. It doesn't matter whether we consider a brother–sister mating, a father–daughter mating or a mother–son mating. The principle is the same for all, but it is simplest to consider a brother–sister mating. If both brother and sister are identically heterozygous their offspring will be a highly variable mish-mash of genetic recombinations. This follows from elementary Mendelian genetics and would apply, in principle, to all animals and plants, not just termites. If you take uniformly heterozygous individuals and cross them, either with each other or with one of the homozygous parental strains, all hell breaks loose, genetically speaking. The reason can be looked up in any elementary textbook of genetics and I won't spell it out. From our present point of view, the important consequence is that during this stage of the development of a termite colony, an individual is typically closer, genetically, to its siblings than to its potential offspring. And this, as we saw in the case of the haplodiploid hymenoptera, is a likely precondition for the evolution of altruistically sterile worker castes.

But even where there is no special reason to expect individuals to be clos*er* to their siblings than to their offspring, there is often good reason to expect individuals to be *as close* to their siblings as to their offspring. The only condition necessary for this to be true is some degree of monogamy. In a way, the surprising thing from Hamilton's point of view is that there are not more species in which sterile workers look after their younger brothers and sisters. What *is* widespread, as we are increasingly realizing, is a kind of watered-down version of the sterile worker phenomenon, known as 'helping at the nest'. In many species of birds and mammals, young adults, before moving out to start families of their own, remain with their parents for a season or two and help to rear their younger brothers and sisters. Copies of genes for doing this are passed on in the bodies of the brothers and sisters. Assuming that the beneficiaries are full (rather than half) brothers and sisters, each ounce of food invested in a sibling brings back just the same return on investment, genetically speaking, as it would if invested in a child. But that is